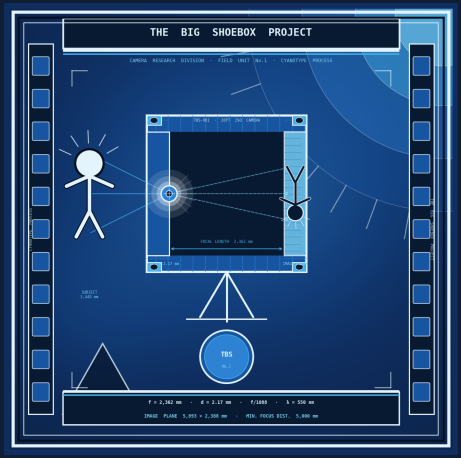


The Big Shoebox Project

TBS-002 — A Pinhole Camera for the Classroom

Cardboard Box Camera · Ø0.794mm Pinhole · f/576 · 457mm Focal Length

Generated November 13, 2026

2 chapters · thebigshoebox.org

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For Teachers — At a Glance

Everything you need to decide whether this fits your class. The full teaching guide and build follow.

What students make	A cyanotype (Prussian-blue) pinhole photograph — a real lensless camera image, developed in nothing but plain water.
Ages	K–5 through college. The build and chemistry scale with the grade; for younger classes the teacher handles all chemistry and students compose, expose, and wash.
Time	Build once (~1–2 hours; the camera is reusable for years). Each printing session is ~30 min hands-on + 10–40 min unattended exposure + ~15 min wash — one class period plus an exposure window.
Group size	2–4 students per camera; a class set is one or two cameras run in rotation.
Space & tools	A normal classroom, a sunny outdoor spot, and a sink. No darkroom required.
Cost	~$108–165 for the camera and the first ~20 prints; about $6–10 per print in consumables thereafter.
Curriculum	Physics (light & optics) · Chemistry (an iron redox reaction) · Math (pinhole-size & exposure formulas) · Art & history (composition; the 1842 “blueprint”).

Safety at a glance

Cyanotype is among the lowest-hazard photographic chemistries. The teacher weighs and dissolves the dry powders wearing nitrile gloves and eye protection; students handle only the diluted working solution, and only from middle school up. Ammonium dichromate — a Category 1A carcinogen — is left out of the school design entirely. Never mix potassium ferricyanide with acids. The finished, washed print is inert and safe to touch.

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Chapter 1

A Pinhole Camera for the Classroom

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Chapter 2

Shopping List

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